

SWITCHOVER TO IPv6 BY MARCH



Yes, the techies have done it again—brought the world to the brink of another disaster, because they failed to foresee the exponential growth of the Internet! Read on to learn about why IPv4 shows threatening signs of becoming the next Y2K disaster

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Internet has become an essential part of our lives. Everybody is on the Net, surfing with different types of devices including PCs, laptops, iPads and mobile phones. Each of these needs a unique IP address on the Internet, when directly connected. The present Internet Protocol (IP) addressing system—IPv4—uses a 32-bit number, represented in 4 octets, for example, 192.16.1.2.

The problem

Since IPv4 addresses are 32-bit numbers, there are only 2^{32} , or just over four billion possible addresses available; and they have all been allocated. On

February 3, 2011, the Internet Assigned Numbers Authority (IANA), which controls the worldwide distribution of IP addresses, allocated the final pool of IPv4 address space to the five Regional Internet Registries (RIRs). This means that there are no longer any IPv4 addresses available for allocation from the IANA to the five RIRs.

So does this mean the end of the Net? Well, not really. Luckily, this problem was foreseen and an alternative to IPv4 developed. The new addressing scheme, IPv6, uses 128-bit numbers for addresses. Hence, there are 2^{128} individual addresses available (approximately 3.4×10^{38}). All new IP addresses from now on will be IPv6.

What then is the 'next Y2K'? The

problem is that IPv4 is not directly compatible with IPv6. This means that devices connected via IPv4 (which make up most of the existing Internet) cannot communicate directly with systems connected using IPv6, and vice versa. If your website is on an IPv4 connection, it won't get traffic from IPv6 sites/users, and the reverse is also true.

Getting traffic is essential to any business on the Net; nobody can afford to lose IPv6 users. Hence, to continue offering customers the same level of service, every organisation needs to be planning IPv6 deployment now.

The IPv6 compliance scenario

All major US government federal agencies have migrated their infrastructure (network backbones) to IPv6. All agency networks were enabled to interface with this infrastructure with effect from June 2008. The US government's chief information officers require that all new IT procurements be IPv6-compliant. An IPv6-compliant product or system must be able to receive, process, and transmit or forward (as appropriate) IPv6 packets, and should inter-operate with other systems and protocols in both IPv4 and IPv6 modes of operation.

The US Defence Department, with its \$30 billion budget, has been buying only IPv6-compliant networking gear since October 2003, and had already achieved full IPv6 compliance in 2008. ICANN, the company that handles the domain names and IP addresses for the world, has also enabled its servers to be IPv6-compliant. There has been significant IPv6 adoption in Japan and Korea, with production networks and consumers paying for IPv6-based services. China is spending millions developing a new backbone network that is reportedly going to be IPv6. The European Union (EU) has also heavily financed research and development in IPv6 backbone networks and innovative services that leverage many of the beneficial features of IPv6.